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VIHAR

K J Somaiya Institute of Technology
(Formerly known as K J Somaiya Institute of Engineering and Information Technology)
An Autonomous Institute Permanently Affiliated to University of Mumbai.

TRUST

Department of IT

Cover page ASSIGNMENT Academic Year: 2026-27 Class: SY Sem: III

Subject: DSA Course Code:

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ASSIGNMENT NO : 1/2 Assignment - II		Date of Performance: 23.10.2026	Date of Correction: 23.10.2026
Rubrics	Marks (20)	Grade	Signature of Faculty
Participation and Involvement (5)	4	17 E	
Preparation and Knowledge (5)	4		
Accuracy and neatness of Documentation(5)	5		
Punctuality (5)	4		

Grades: Excellent – E (17 – 20)
Very Good – A (13-16)
Average – B (9-12)
Satisfactory – C (5-8)
Unsatisfactory – D (< - 5)

Assignment - II

1] Develop & evaluate a student recommendation system in C/C++ using appropriate sorting & searching algorithm.

↳
```
#include <iostream>
# include <string>
# include <vector>
# include <chrome>
#include <studio>
# include <algorithm>

using namespace std;
using namespace std; chrome;

struct student {
        int id;
        string name;
        float score;
     };
void bubble sort (vector < student >& arr){
    int n= arr.size();
for (int i=0; i<n-1; i++)
 {
 for (int j=0; j<n-i-1 ; j++)
 {
 comparison count++;

If (arr[j] . score < arr [j+1] . score)
   {
        swap (arr[j] . arr [j+1]);
 }}}}
```

```
void selection sort (vector student > &arr)
 {
   int n = arr size();
for (int i=0 ; i<n-1 ; i++) {
     int Max Idx = i;
 for (int j= i+1 ; j<n ; j++) {
comparisons count ++;

if (arr[j]. score > arr [max idx]. score) {
    max idx = j;
}}
 swap( arr[i] , arr [maxIdx]);
 }
}

void insertion sort (vector <student> &arr)
{
 int n=arr.size();
   for (int i=1 ; i>n ; i++)
  {
    student key = arr[i])
        int j = i-1;
 while( j>=0) {
        comparison count++;
   if ( arr[j]. sore < key.score) {
              arr [j+1] = arr[j];
                   j--;
        } else {
            break;
        }}
           arr[j+1] = key;
                       }}
```

```
void merge(vector<student>&arr, int left,
 int mid, int right)
 {
     int n1 = mid - left +1;
      int n2= right -mid;
  vector < student > L(n1), R(n2);

 for(int i=0 ; i<n1; i++) L[i] = arr[left+i]
 for (int j=0  ; j<n2 ; j++ ) R[j] =
                              arr[mid+1+j];

  int i=0, j=0, k=left;
      while( i<n1 && j<n2){
          comparison count ++;
   if ( L[i].score >= R[j].score){
          arr[k++] = L[i++];
       } else{
          arr[k++]= R[j++];
       }}
  while (i<n1) arr[k++] = L[i++]
  while [j<n2] arr[k++] = R[j++];

 void reset counter(){
         comparison count=0;
     }

 void quick sort(vector<student>&arr,
  int low, int high)
  { if (low<high){
            int pi= partition (arr, low, high));
        quick sort (arr, low, pi-1);
        quick sort (arr, pi+1, high);
                                 }}
```

```
int Binary search( count vector<student >
    ,& arr, float target score) {

int low=0 ; high = arr.size()-1;
   while (low< =high) {

comparison count ++
   int mia= low + (high-low)/2;

if (arr[mid]).score = target score return mid;
if (arr[mid]). score < target score) high = mid-1

else low = mid+1;
}
return -1;
 }

vector<student > generate dataset (int size){
   vector <student> list (size);
  for (int=0 ; i< size ; i++){

  list[I].name = "Student" + + string (i+1);
  list[i]. score = ( rand ()%1000} /10.0;
 }
  return list;
}
int main(){
        arrange (42);
  count << "Student Recommendation system
bench searching .
   return 0;
}
```